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Dynamic switching between brain networks predicts creative ability

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Qunlin Chen [1,2,9], Yoed N. Kenett [3,9] ✉, Zaixu Cui [4], Hikaru Takeuchi [5], Andreas Fink [6], Mathias Benedek [6], Daniel C. Zeitlen [2], Kaixiang Zhuang [7], James Lloyd-Cox[8], Ryuta Kawashima [5], Jiang Qiu [1] ✉ & Roger E. Beaty [2]

Creativity is hypothesized to arise from a mental state which balances spontaneous thought and cognitive control, corresponding to functional connectivity between the brain's Default Mode (DMN) and Executive Control (ECN) Networks. Here, we conduct a large-scale, multi-center examination of this hypothesis. Employing a meta-analytic network neuroscience approach, we analyze resting-state fMRI and creative task performance across 10 independent samples from Austria, Canada, China, Japan, and the United States ($N = 2433$)—constituting the largest and most ethnically diverse creativity neuroscience study to date. Using time-resolved network analysis, we investigate the relationship between creativity (i.e., divergent thinking ability) and dynamic switching between DMN and ECN. We find that creativity, but not general intelligence, can be reliably predicted by the number of DMN-ECN switches. Importantly, we identify an inverted-U relationship between creativity and the degree of balance between DMN-ECN switching, suggesting that optimal creative performance requires balanced brain network dynamics. Furthermore, an independent task-fMRI validation study ($N = 31$) demonstrates higher DMN-ECN switching during creative idea generation (compared to a control condition) and replicates the inverted-U relationship. Therefore, we provide robust evidence across multi-center datasets that creativity is tied to the capacity to dynamically switch between brain networks supporting spontaneous and controlled cognition.

Creative thinking is a critical human capacity, enabling the progress of civilization through continuous innovation in diverse scientific and artistic domains. In recent decades, behavioral and neuroimaging studies have yielded a considerable understanding of the neural and cognitive mechanisms underlying how people generate novel and useful (i.e., creative) ideas[1–3]. Growing evidence suggests that creative thought emerges from the interaction between spontaneous associations and deliberate, cognitive control processes, primarily driven by interactions between the Default Mode Network (DMN) and various other networks[4–6]. In particular, greater creative thinking has been linked to the coupling of the DMN and the Executive Control Network (ECN).

The DMN consists of cortical midline (including the medial prefrontal cortex and posterior cingulate cortex), posterior inferior parietal regions, and the medial temporal lobe; this brain network is known to involve internally directed or self-generated thought, such as the spontaneous activation of memories, mind wandering, semantic integration, associative idle thoughts, and mental simulation—cognitive processes thought to support creative idea generation[7–14]. The ECN consists of lateral prefrontal and anterior inferior parietal regions, and when coupled with the DMN, supports controlled semantic retrieval and cognitive flexibility, which benefits the selection and evaluation of creative ideas[15–20]. Numerous neuroimaging studies have consistently found that creative thinking recruits regions within the DMN and ECN[8,21–26]. In addition, an individual creative thinking ability can be predicted by functional interactions within and between the DMN and ECN both at rest and during creative task performance using static networks methods[27–34]. For example, functional

[1]Faculty of Psychology, Southwest University, Chongqing, China. [2]Department of Psychology, Pennsylvania State University, University Park, Pennsylvania, USA. [3]Faculty of Data and Decision Sciences, Technion-Israel Institute of Technology, Haifa, Israel. [4]Chinese Institute for Brain Research, Beijing, China. [5]Division of Developmental Cognitive Neuroscience, Institute of Development, Aging and Cancer, Tohoku University, Sendai, Japan. [6]Department of Psychology, University of Graz, Graz, Austria. [7]IInstitute of Science and Technology for Brain-Inspired Intelligence, Fudan University, Shanghai, China. [8]Department of Psychology, Goldsmiths, University of London, London, UK. [9]These authors contributed equally: Qunlin Chen, Yoed N. Kenett. ✉e-mail: yoedk@technion.ac.il; qiuj318@swu.edu.cn

connectivity features associated with the default mode, executive control, and salience networks can predict individual creative performance during divergent thinking tasks, and these features can be generalized across several independent resting-state fMRI samples[27]. Given the roles of the DMN and ECN in spontaneous and controlled cognition, respectively, DMN-ECN connectivity is thought to reflect a coordination between spontaneous/generative and controlled/evaluative processes[5,35,36]. However, static network approaches fail to capture the dynamic coordination of cognitive states related to reconfiguration and interactions between networks[25,32,37,38]. Therefore, dynamic reconfiguration-based functional analyses are needed to identify specific interaction patterns between the DMN and ECN related to creative ability.

How does the creative brain achieve dynamic coordination between spontaneity and control? According to the dynamic framework of spontaneous thought[36,39], creativity is a mental state that requires an optimal balance between spontaneous thought generation and goal-directed processing—corresponding to a balanced state of DMN and ECN communication. Dual-process theories of creativity also emphasize that creative cognition involves two different phases—idea generation and evaluation—dominated by cooperation between large-scale functional brain networks including the DMN and ECN[1,2,25,35,38]. Therefore, one possibility is that higher creative people can flexibly switch between segregation and integration of DMN and ECN—oscillating between states where the brain networks operate independently and interdependently, respectively—which may reflect an optimal balance or "sweet spot" between the networks that leverage their independent and collective capacities. Despite the intuitive appeal of this theory, very little empirical evidence to support this claim exists. Several studies based on stages of creative processing have pointed out that the DMN predominantly acts on the generative phase in the early stages of creative production, whereas increased coupling between the DMN and ECN underlies the evaluative phase in the later stage[21,34,38], implying that the ECN may monitor and guide spontaneous associative/generative processes[25,40–42]. Additionally, the temporal connectivity of large-scale brain networks based on resting-state fMRI data supports the dynamic functional configuration associated with creative cognition[32,43]. However, the precise interaction patterns and the mechanistic role of this DMN-ECN coupling in creative thinking remain unclear.

This dynamic reconfiguration of brain network connectivity can be divided into two distinct patterns or "states" that the brain routinely occupies: segregation and integration. The segregated pattern corresponds to independent processing within networks, whereas the integrated pattern corresponds to cooperation between networks[44]. Previous research has reported the importance of brain network segregation and integration for performance on diverse cognitive tasks[45], although no prior work has examined such brain network dynamics in relation to creativity. Several studies have, however, observed variations in temporal cooperation between functional networks during creative cognition, suggesting the existence of a possible dynamic transition between functional segregation and integration[32,34]. From the perspective of individual differences, highly creative individuals are more likely to engage in internally directed cognition during rest, which is characterized by intrinsic functional networks and their interactions[1,46,47]. Given that resting-state functional networks shape cognitive task activations[48], individuals with higher creative thinking abilities——potentially conducive to a more efficient cycle of generation and evaluation—may also exhibit altered dynamics of large-scale brain networks (namely, the DMN and ECN) at rest[49,50]. Studies have consistently shown that intrinsic network dynamics during rest reflect fundamental organizational properties that predict individual differences in cognitive abilities[45,51–53]. The pattern and frequency of network switching between segregated and integrated states appear particularly important, as it indicates the brain's capacity to flexibly coordinate different cognitive processes[53–56]. Examining these dynamics during rest allows us to characterize stable individual differences in network organization across large samples, complementing and extending insights from task-based studies of creative cognition. This approach is especially valuable given evidence that creative ability relates to intrinsic patterns of brain network interaction that persist across both rest and task states. Consistent with this view, dynamic connectivity studies have reported that the temporal variability of resting-state functional connectivity among regions within DMN and ECN was positively related to creative thinking ability[43,57]. Further, a recent study provided a more nuanced understanding of the neurocognitive mechanisms of the creative brain, indicating that generative and evaluative stages may involve different combinations of associative and controlled processes, and thus recruit different proportions of DMN and ECN activity[5].

Independently from creative thinking, more flexible brain network reconfiguration (that is, highly frequent brain state switching) may reflect a functional balance between segregated and integrated states that helps to foster cognitive abilities such as memory retrieval[45,53,58]. For example, a recent investigation reported that while stronger whole-brain segregation supports crystallized intelligence and processing speed and integration fosters better general cognitive ability, a balance between segregation and integration is associated with better memory[53]. By contrast, hyper-connectivity (i.e., increased positive connectivity) between the DMN and the ECN, or other networks, is associated with rigid behaviors and thoughts linked to certain psychiatric disorders (e.g., autism spectrum disorder), while hypo-connectivity (i.e., decreased positive connectivity) is associated with impairments in executive function and social cognition[58–60]. Returning to the present study, no research to date has investigated dynamic interactions between the DMN and ECN in creative thinking regarding which patterns of switching between segregated and integrated states can benefit individual creativity.

In the present study, we investigate whether individual differences in creative ability can be reliably predicted by the dynamic interaction between the DMN and ECN. To this end, we analyze large-scale, multi-center datasets including structural MRI, resting-state fMRI, and creative ability data (assessed by divergent thinking tests administered outside the scanner) from five centers in Austria, Canada, China, Japan, and the United States ($N = 2433$)—constituting (to our knowledge) the largest and most ethnically diverse brain-based analysis of creativity to date. We applied a meta-analytic network neuroscience approach to integrate the findings by conducting a new analysis using the raw data from each dataset. Here, two dichotomous neural states, segregated and integrated functional connectivity patterns between the DMN and ECN, were determined using time-resolved analyses of resting-state fMRI data. Building on previous research[5,32,43], we hypothesized that a higher frequency of switching between DMN-ECN segregation and integration over time will predict better creative performance. We also hypothesized that higher creative performance may benefit from a functional balance between segregation and integration in these brain networks. Finally, to provide a more direct validation of the neural coupling dynamics, we conducted an external validation analysis using task-based fMRI data in an independent sample of Chinese adults ($N = 31$) and tested whether the brain network dynamics assessed during rest in the full sample extend to active performance during a creative thinking task.

Results

Sample composition and behavioral findings

Ten datasets spanning five centers were analyzed, including data from 2433 healthy individuals aged from 16 to 58. Figure 1A summarizes the sample sizes and age ranges of each dataset (Supplementary Table 1) and Fig. 1B describes the data distribution of creative performance of each dataset. Individual creative performance was assessed through the widely used Alternate Uses Task[61], where participants generate creative uses to objects and their responses were scored by trained 2-4 raters (see Supplementary Dataset Information for more detail). Behavioral analyses (Supplementary Table 2) showed no consistent gender differences across the 10 datasets, which is in line with previous literature. No significant correlation was found between creativity scores and age except for one dataset (UG_S1). Thus, the absence of consistent findings across diverse datasets suggests that gender differences in creativity scores and the associations between creative performance and age lack robustness due to sample heterogeneity and the very

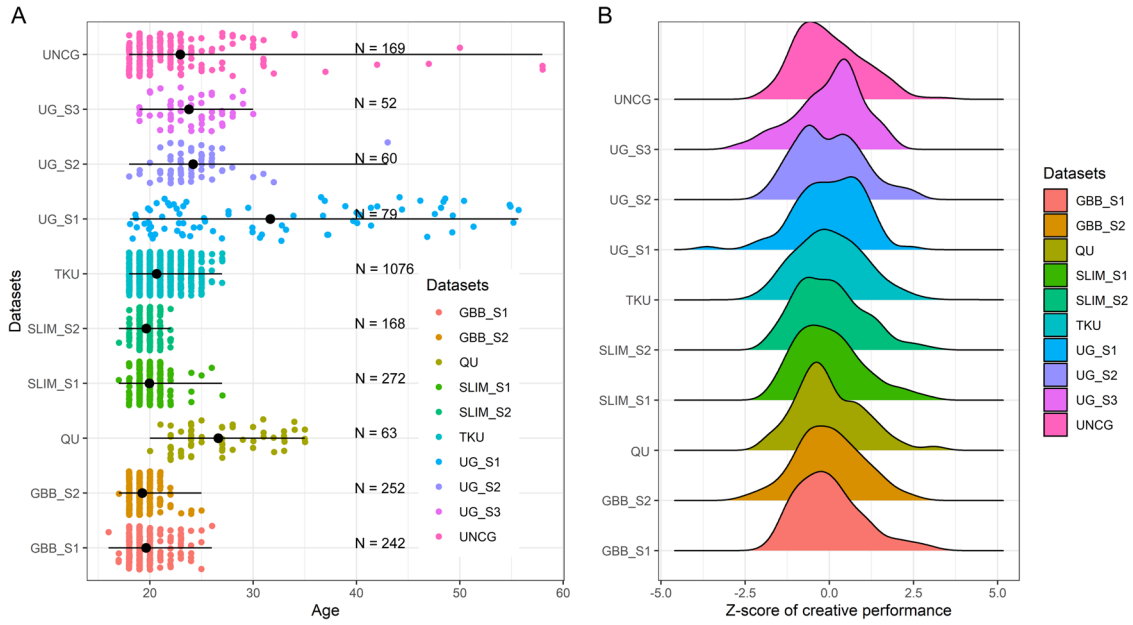

Fig. 1 | Datasets characteristics. A The total number of participants and the age ranges of each dataset. The position of the black dot on each line indicates the median age of that dataset. **B** The distribution of creative performance based on human-rated originality scores on divergent thinking tasks in each dataset.

positively skewed distribution in age. Among six datasets containing both intelligence and creativity scores, three datasets showed a significant correlation between the two constructs: UNCG, $r = 0.32$, 95% CI [0.17–0.46]; UG_S1, $r = 0.33$, 95% CI [0.12–0.51]; UG_S3, $r = 0.39$, 95% CI [0.13–0.60]; SLIM_S1, $r = 0.07$, 95% CI [−0.05 to 0.19]; GBB_S1, $r = 0.13$, 95% CI [−0.04 to 0.30]; and GBB_S2, $r = -0.01$, 95% CI [−0.14 to 0.12], consistent with past work demonstrating positive associations between intelligence and creativity[62,63].

The relationship between network switching frequency and creative performance

Next, we turned to test our primary hypothesis regarding whether creative thinking benefits from a tendency to dynamically switch between segregation and integration of the DMN and ECN. The switching frequency was calculated by the number of transitions between segregated and integrated states across the time windows (see "Methods"). Pearson correlation analyses confirmed our hypothesis across 7 of the 10 datasets (Fig. 2): the switching frequency of DMN-ECN segregation and integration was significantly and positively associated with creative performance. Next, we applied a meta-analytic approach to calculate the overall weighted effect size between DMN-ECN state switching frequency and creative ability across the ten datasets. The overall effect size was significant applying random effects at the dataset level, $g = 0.174$, 95% CI [0.08, 0.27], $Z = 3.69$, $p < 0.001$ (Fig. 3). The Q test for heterogeneity was significant, $Q(9) = 20.67$, $p = 0.014$, $I^2 = 56.45\%$, indicating the necessity for exploring potential moderators of these effects[64].

Considering that the three moderating variables—location (corresponding to five centers), scanner type, and scoring methods are almost confounded, we conducted a moderator analysis that included all three variables as moderators together. Additionally, following the subgroup analyses, we calculated the pure effect of each group after excluding other variables, such as location and scanner type. The following moderator analyses revealed location, scanner type, and scoring methods, $QB(5) = 27.05$, $p < 0.001$, significantly and jointly moderated the overall effect. Subsequently, we performed subgroup analyses within center, scanner type, and scoring methods for creativity in which the number of effects (k) is at least 3: Austria datasets (k = 3) have a significant and medium positive effect, $g = 0.311$, 95% CI [0.17, 0.46], $Z = 4.19$, $p < 0.001$, and China datasets (k = 4) have a relatively weak and significant positive effect, $g = 0.114$, 95% CI [0.01, 0.22], $Z = 2.05$, $p = 0.04$. Datasets with the magnetome MRI scanner (k = 4) have a significant and medium positive effect, $g = 0.284$, 95% CI [0.18, 0.39], $Z = 5.30$, $p < 0.001$, whereas datasets with the Trio MRI (k = 5) have a significant and weak positive effect, $g = 0.11$, 95% CI [0.01, 0.20], $Z = 2.08$, $p = 0.037$. Datasets with scoring using originality alone (k = 7) have a significant and medium positive effect, $g = 0.18$, 95% CI [0.09, 0.28], $Z = 3.74$, $p < 0.001$, and datasets with scoring using the sum of originality and fluency (k = 3) have a weak but significant effect, $g = 0.14$, 95% CI [0.01, 0.27], $Z = 2.04$, $p = 0.042$. In addition, sample size, $QB(1) = 2.66$, $p = 0.103$, age, $QB(1) = 3.66$, $p = 0.06$, scanning time, $QB(1) = 0.85$, $p = 0.36$, and gender ratio of sample, $QB(1) = 1.36$, $p = 0.244$, did not significantly moderate the overall effect, indicating that data origin (including the locations and MRI and scoring protocol) but not participant-related variables influenced the overall effect. Thus, more creative individuals tend to switch more frequently between segregation and integration of the DMN and ECN at rest.

Furthermore, the results remained statistically significant when using the robust correlation method, which does not rely on assumptions of normality or linearity and can handle outliers better than traditional correlation methods. Moreover, the overall effect size was found to be significant, $g = 0.149$, 95% CI [0.07, 0.23], $z = 3.47$, $p < 0.001$, using random effects model (Supplementary Fig. 1). The Q test for heterogeneity was not significant here, $Q(9) = 16.08$, $p = 0.065$, $I^2 = 44.03\%$, indicating relatively low variability among the studies.

The relationship between switching frequency and intelligence

Given the conditional relationship between creativity and intelligence, we applied a meta-analytic approach to determine whether intelligence, regarded as a control task, also relates to segregation and integration of DMN and ECN, as we expected (and found) for creativity. Thus, the same analyses were performed among 6 datasets ($N = 908$, see Supplementary Table 3) which included intelligence scores in addition to creativity scores. The overall effect size was not significant, $g = 0.023$, 95% CI [−0.042, 0.089], $Z = 0.699$, $p = 0.485$ with no significant heterogeneity, $Q(5) = 4.076$, $p = 0.539$. In contrast, the overall effect size for the relationship between switching frequency and creative performance was significant among the same 6 datasets, $g = 0.20$, 95% CI [0.05, 0.36], $z = 2.54$, $p = 0.01$, $Q(5) = 15.05$, $p = 0.01$. The parameter estimate difference between the two models (Intelligence v.s. Creativity) is significant, $z = -2.069$, $p = 0.038$, indicating

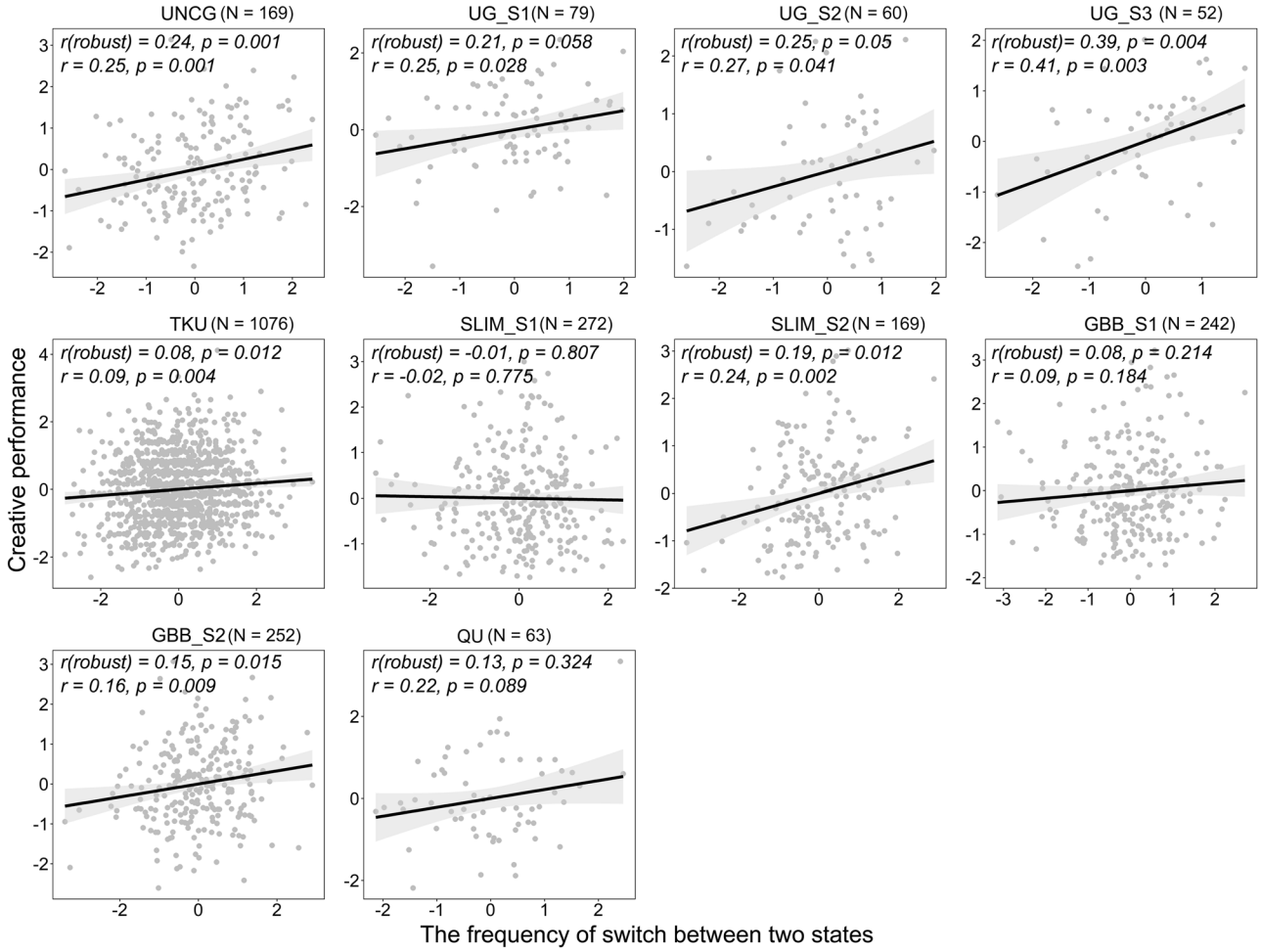

Fig. 2 | Associations between creative performance and the switching frequency of DMN-ECN segregation and integration. The *r* (robust) indicated the value of robust correlation in each dataset. The fitting regression line is shown in black, and the 95% confidence interval for the line is shown in gray shading. Both the *x* (creative performance) and *y* (creative performance) axes are standardized value after adjusting for gender, age, mean framewise displacement, and global mean signal.

that the DMN-ECN switching effect is specific to creative ability and does not extend to intelligence.

Creative performance and the balance of DMN-ECN segregation and integration

Previous empirical evidence has shown that individuals with various degrees of functional balance between network segregation and integration demonstrated differential cognitive ability[45,65]. Furthermore, frequent switching between segregated and integrated states results in a functional balance, and optimal balance (i.e., when the magnitude of balance is equal to zero) might reflect maximum flexibility in dynamic transitions between the two states[53]. Here, we aimed to test whether there is a nonlinear relationship between creative performance and DMN-ECN segregation and integration balance; we therefore performed an ANOVA test and compared AIC values between the linear and quadratic regression model in each dataset.

We found that the quadratic regression model was significantly better fitting than the linear regression model (Fig. 4) in three datasets: UG_S1 ($AIC_{lm} = 229.93$, $AIC_{qm} = 223.4$; $F(1, 76) = 8.66$, $p = 0.004$), TKU ($AIC_{lm} = 3059.44$, $AIC_{qm} = 3051.53$; $F(1, 1073) = 9.94$, $p = 0.002$), and SLIM_S2 ($AIC_{lm} = 482.41$, $AIC_{qm} = 479.78$; $F(1, 165) = 4.61$, $p = 0.033$). There was no significant difference between the two models in the remaining seven datasets (Supplementary Table 4).

To summarize the effect sizes of the quadratic regression model across all datasets, the partial correlation coefficients of the quadratic term in each regression model were meta-analyzed using a random-effects model with restricted maximum likelihood. The overall effect size was small but

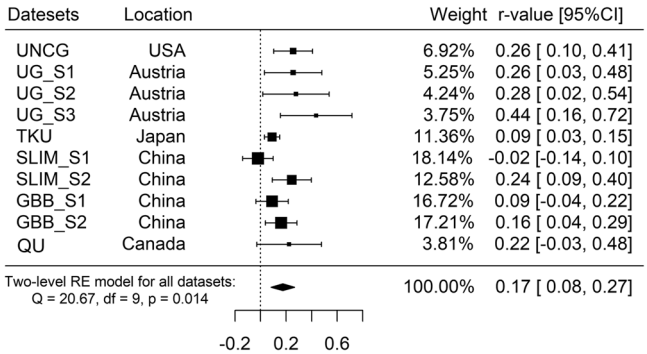

Fig. 3 | Forest plots of ten effect sizes included in the meta-analysis. Forest plot of effect sizes for the correlation between creative performance and switching frequency (calculated using resting-state fMRI data). On the left side, forest plots display the name and location of each dataset; on the left of dashed vertical line (representing no effect), the forest plots provide the weight, the observed effect, and the 95% confidence interval, in which the size of a square is proportional to the weights determined by the inverse of datasets' variance; the bottom line shows the overall effect size (right) and heterogeneity test (left).

significant, $g = -0.07$, 95% CI $[-0.14, -0.01]$, $z = 2.26$, $p = 0.024$, indicating that moderate cooperation—neither more nor less—between the DMN and ECN may be conducive to generating creative ideas. This finding may suggest that across multi-center datasets, creative ability benefits from an

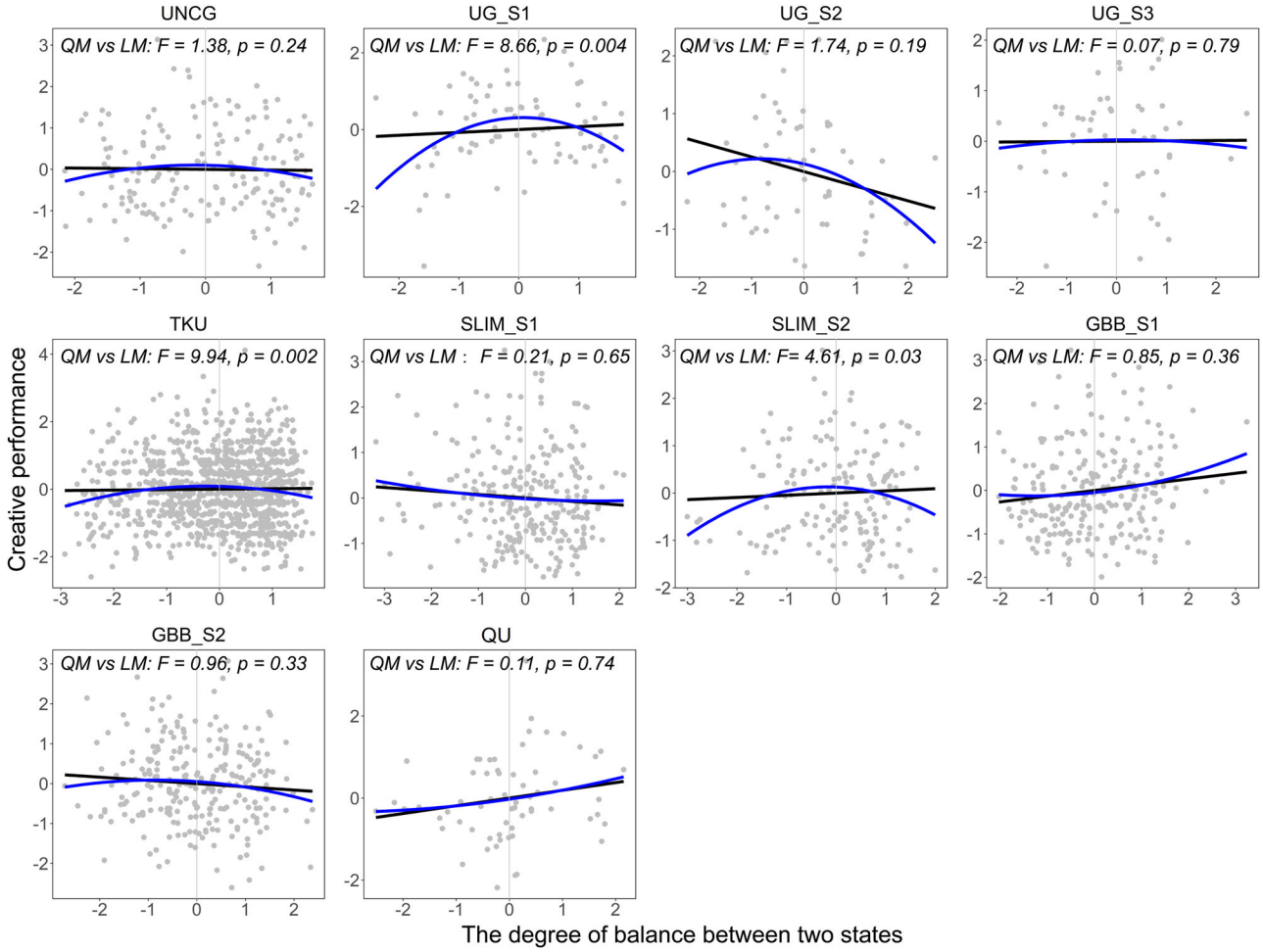

Fig. 4 | Associations between creative performance and the balance of DMN-ECN segregation and integration. Linear (black fitted line) and quadratic relationship (blue fitted line) of creative performance (*y*-axis) and DMN-ECN segregation and integration balance (*x*-axis), and with the comparison between linear fitted model and quadratic fitted model for each dataset (calculated using resting-state fMRI data). Both the *x* and *y* axes are standardized value after adjusting for gender, age, mean framewise displacement, and global mean signal.

optimal balance between spontaneous and controlled processes, supported by moderate switching between DMN-ECN segregation and integration. The Q-test statistic, $Q(9) = 16.72$, $p = 0.053$, $I^2 = 48.81\%$, suggests that little heterogeneity exists.

Reproducibility

Importantly, we evaluated the reproducibility of our findings using mega-analyses, an approach that pools all available individual-level data from various datasets. This method allows us to exclude non-relevant covariates, such as demographic (such as gender and age) and location factors, and focus on specific effects within a single model. The results of the mega-analysis are summarized in Supplementary Table 5, indicating that individuals with higher switching frequency of DMN-ECN segregation and integration exhibited higher creative performance, $\beta = 0.15$, $t(2419) = 4.6$, $p < 0.001$ (Fig. 5A). By contrasting the linear and nonlinear models using mega-analysis, the results showed that balance of dynamic switching between DMN-ECN segregation and integration, as the quadratic terms, can significantly predict the creative performance, $\beta = -0.06$, $t(2418) = -3.81$, $p < 0.001$ (Fig. 5B), and the nonlinear model ($AIC = 6925.45$, Supplementary Table 6) is better than the linear model ($AIC = 6931.19$; $\beta = 0.01$, $t(2419) = 0.44$, $p = 0.658$, Supplementary Table 7) that only includes the linear term (the balance of dynamic brain states) as a fixed factor, $\chi^2 = 7.74$, $p = 0.005$. Overall, the mega-analysis yielded results similar to those of the meta-analytic approach. Reanalyzing data using the Shen brain atlas[66], which consists of 54 regions of interest within DMN and ECN also confirmed our results: the overall effect size was significant using the random-effects model, $g = 0.09$, 95% CI [0.01, 0.16], $Z = 2.18$, $p = 0.029$ (Supplementary Fig. 2).

External validation using task-based fMRI data

We identified dynamic interactive patterns between the DMN and ECN associated with individual creative performance in a large-scale, multi-center, resting-state fMRI dataset. However, can these patterns be validated in the context of a divergent thinking task, which involves creative cognitive processes, as opposed to a control task that only requires common cognitions (e.g., memory retrieval)? To address this, we conducted an external validation analysis using task fMRI data, with thirty-one participants completing both the Alternate Uses Task (AUT), requiring production of creative object uses, and the Object Characteristics Task (OCT), requiring production of common object characteristics, during fMRI scanning. The switching frequency and the degree of balance between two states (i.e., DMN-ECN segregated or integrated state) were derived from task-relevant events. Specifically, we extracted these two dynamic indices by precisely pinpointing AUT and OCT trials when subjects initiated thinking, generating up to four ideations (Fig. 6A). The two indices for each task were then computed for each participant by calculating the switch frequency and balance of DMN-ECN segregation and integration in each run.

The results revealed that AUT (0.14 ± 0.05; Fig. 6B) recruited a significantly higher switching frequency than OCT (0.12 ± 0.07; Fig. 6D), $t(86) = 2.59$, $p = 0.011$, Cohen's $d = 0.28$ (Fig. 6C). This suggests that creative

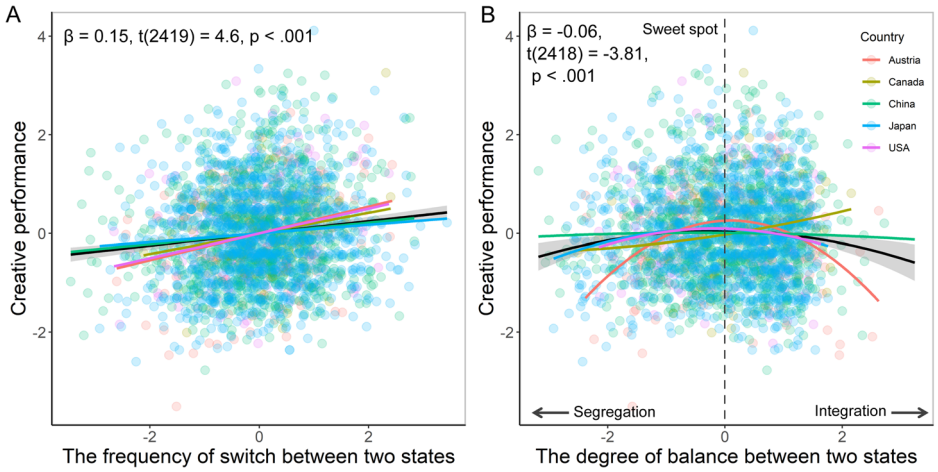

Fig. 5 | Mixed effect models. A The linear relationship between creative performance and switching frequency between segregated and integrated DMN-ECN states (calculated using resting-state fMRI data); the black line marks the overall trend, and the 95% confidence interval for the line is shown in gray shading. **B** The quadratic relationship between creativity score and functional balance between segregated and integrated DMN-ECN states (calculated using resting-state fMRI data); the black line represents the overall trend, with the 95% confidence interval indicated by the gray shading, while the colored lines depict trends for each center. The left side of the *x*-axis indicates extreme segregation, the right side indicates extreme integration, and the dotted line in the center represents the optimal balance or "sweet spot".

thinking may be supported by interactions between the DMN and ECN compared to general cognitive processes, such as extracting concepts from existing knowledge. A linear mixed-effects model did not show that AUT performance (originality score of ideas) was significantly associated with the switching frequency of DMN-ECN segregation and integration during creative thinking, $\beta = 0.02$, $t(63) = 0.04$, $p = 0.971$, Cohen's $d = 0.01$ (Fig. 6E); similarly, OCT performance was not related to the switching frequency, $\beta = -0.24$, $t(59) = -0.82$, $p = 0.418$, $d = -0.21$ (Fig. 6F). However, a nonlinear mixed-effects model analyses showed that the degree of balance between DMN-ECN segregation and integration is related to AUT performance, $\beta = -0.41$, $t(57) = -2.62$, $p = 0.011$, $d = -0.69$, and the nonlinear model (AIC = 53.82) is better than the linear model (AIC = 58.46, $\beta = -0.12$, $t(55) = -1.36$, $p = 0.178$, $d = -0.37$) that only includes the linear term (the balance of dynamic DMN-ECN states) as a fixed factor, $\chi 2 = 6.64$, $p = 0.01$ (Fig. 6G). But the balance of dynamic DMN-ECN switching is not related to OCT performance in both the nonlinear model, $\beta = 0.13$, $t(54) = 1.45$, $p = 0.152$, $d = 0.39$, and the linear model, $\beta = 0.03$, $t(57) = 0.74$, $p = 0.465$, $d = 0.19$ (Fig. 6H).

Thus, the task-based fMRI findings validate the resting-state results: creative thinking elicits greater dynamic switching and more balanced interactions between default mode and executive control networks compared to non-creative cognition, thereby supporting individual creative performance.

Discussion

Our meta-analytic analyses revealed that creative performance can be reliably predicted from the frequency of switches between the segregated and integrated states of the DMN and ECN at rest: people who switch between these states more often produce more original ideas. Furthermore, high creative thinking ability was associated with a balance between the segregated and integrated states because frequent transitions between these states provided a means to maintain this balance. The convergence between these resting-state patterns across a large sample and their selective emergence during creative task performance (compared to control tasks) suggests these network dynamics represent a specific marker of creative ability. These findings advance our understanding of the neural mechanisms of creativity by identifying specific patterns of network interaction that characterize both the capacity for and active engagement in creative thought.

While much progress has been made, our understanding of the neural bases of creativity remains far from complete. Creativity is a complex construct that recruits both creativity-specific and ordinary cognitive processes such as memory, attention, and cognitive control[41,67,68]. A dynamic network-based approach is particularly well-suited to address the complex interplay of neural mechanisms that engage the whole brain during creative thinking[1]. In this perspective, the present study focused on the entire ECN and DMN in creative cognition, while disregarding differences between specific regions and subnetworks within these large brain networks. In addition, we expected to find domain-general patterns of interactive DMN-ECN connectivity related to creative cognition. Indeed, in the present research, we found no significant correlation between intelligence and DMN-ECN switching frequency across six datasets. Additionally, we observed a significantly lower DMN-ECN switching frequency under the control (memory retrieval) task compared to the divergent thinking task. This observation may suggest that these dynamic interactions between the DMN and ECN characterize creative cognition in particular, rather than other cognitive abilities.

A body of studies have shown that creative thinking is generally dependent on the capacity of the DMN to synchronize with other brain networks[25,38], such as the ECN[34], salience network[69], sensorimotor network[70–73], and visual network[74,75]. While coupling between the DMN and ECN has been consistently highlighted as a marker of creativity, such a coupling has been largely shown in "static" functional connectivity research and the role of a specific coupling dynamic as a driver of creativity has so far been largely speculative by analyzing the whole-brain functional connectivity[1,51]. However, few studies have focused on examining the interactions within these networks and how they influence creative cognition[25,38]. Although our research relies on methods involving brain–behavioral phenotype associations, by combining evidence from both resting-state networks with large sample size and cognitive functional networks (i.e., task-based fMRI), our findings further support the idea that functional coupling among regions in the DMN and ECN may reflect a domain-general mechanism of creative cognition. Importantly, a new dynamic interaction pattern between the DMN and ECN was observed in this study, namely that higher creative individuals exhibit optimal neural cooperation between the DMN and ECN. Specifically, the degree of balance between the segregated and integrated states demonstrated an inverted U-shaped relationship with creative performance, where too much (or too little) DMN-ECN coupling may impair creative performance. This finding aligns with previous creativity theories, proposing that creative ideation emerges from the delicate balance between flexibility and stability of cognitive states[76–78]. This equilibrium is crucial for maintaining the optimal combination of novelty and utility throughout the process of creative cognition[76].

Previous research suggested that the DMN may be centrally involved in active recollection of memories and associative processes that yield diverse ideas, whereas the ECN may act to guide associative processes through monitoring, inhibition, and related top-down control processes[26,40,67]. These studies explain the roles of the DMN, ECN, and other networks in creative cognition in two stages: generation and evaluation[2,21,25,38]. Previous creativity theories have suggested that

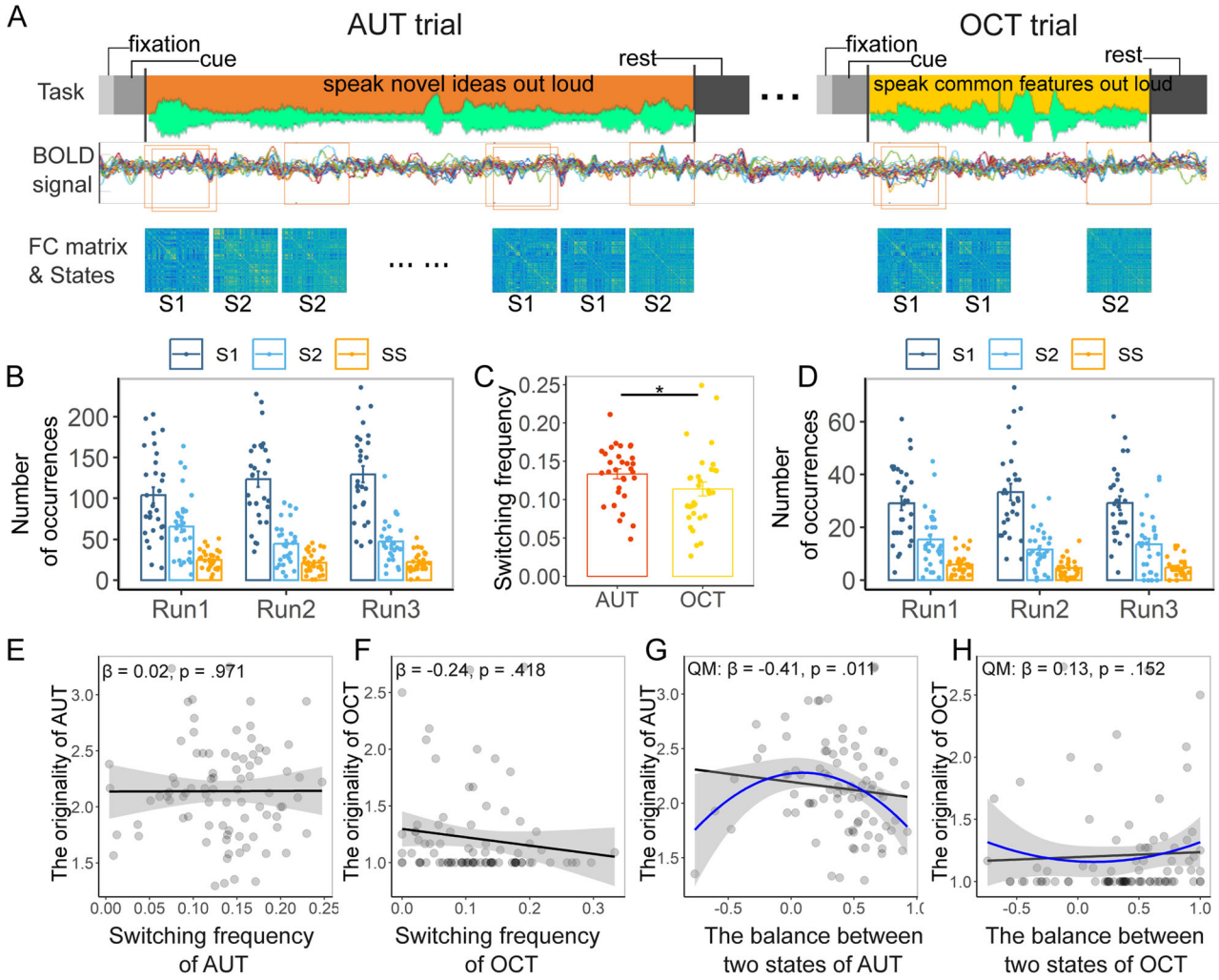

Fig. 6 | fMRI task design, main analysis, and results. A Experimental paradigm of the divergent thinking task, in which participants had to complete an Alternate Uses Task (AUT) and an Object Characteristics Task (OCT) during fMRI scanning. The task paradigm consisted of a fixation cross (1 s), a cue indicating the upcoming condition ("novel uses" or "common characteristics"; 4 s), a thinking period presenting an object cue in text (60 s for AUT; 30 s for OCT). Participants responded by pressing a key and speaking their responses out loud into an MRI-compatible microphone until they either generated four ideas or ran out of thinking time. This was followed by a rest period of at least 6 s, or the remaining time from the thinking phase if applicable. Task fMRI data were preprocessed using fmriprep, and then time courses were extracted from 106 ROIs affiliated with the default and executive control networks derived from Schaefer Atlas; subsequently, a sliding-window approach was used to compute the functional connectivity (FC) matrix, and each matrix was partitioned into segregated (state 1, S1) and integrated (state 2, S2) states using k-means clustering. **B** The left panel shows the mean number of segregated, integrated, and switching states (SS) during the AUT task across three runs. Error bars indicate the standard error of the mean. **C** The average switching frequency for the AUT and OCT task, and paired-sample *t*-tests comparing AUT v.s. OCT (the error bar represents the standard errors) $^*p < 0.05$. **D** The right panel shows the mean number of segregated, integrated states, and switching during the OCT task across three runs. Error bars indicate the standard error of the mean. **E**, **F** The scatterplots show the Pearson correlations between the originality score (i.e., creative performance in the task, *y*-axis) and the switching frequency of DMN-ECN segregation and integration (*x*-axis) in AUT and OCT, respectively. The black line represents the overall trend, with the 95% confidence interval indicated by the gray shading. **G**, **H** Linear (black fitted line, with the 95% confidence interval indicated by the gray shading) and quadratic relationship (i.e., quadratic model, QM; blue fitted line, with the 95% confidence interval indicated by the gray shading) of originality score (*y*-axis) and DMN-ECN segregation and integration balance (*x*-axis), and with the comparison between linear fitted model and quadratic fitted model for AUT and OCT, respectively.

creativity is a continuum[79] and creative outputs occur along a set of cognitive continuums, such as the divergent–convergent thinking continuum, the defocused–focused continuum, the spontaneous–deliberate continuum, the mind-wandering–mindfulness continuum, etc.[80]. In the present neurocognitive framework, such cognitive continuums can map onto the DMN–ECN coupling, in which the DMN and ECN together realize the cycle of generation and evaluation. For example, DMN-ECN coupling might reflect forms of goal-directed memory retrieval and episodic simulation in the stage of idea generation that requires cognitive control[5,81]; whereas similar coupling patterns act to constrain and direct evaluation processes to output ideas that are both novel and appropriate in the stage of idea evaluation[34]. Some evidence suggests that different proportions of DMN and ECN activity within coupling patterns might correspond to generative and evaluative stages of creative thinking[5,21,25,38]. Thus, it is likely that dynamically shifting between generation and evaluation could provide an optimal way to produce a constant stream of creative outputs. Yet, such dynamic switching between generation and evaluation must be carefully balanced, as too much of one without the other is unlikely to produce both novel and appropriate ideas[42,49,50,82].

Our work further—albeit indirectly—establishes a connection between DMN-ECN switching dynamics and the theoretical generation-evaluation creativity model. Direct evidence supporting this hypothesis is needed, and future research should aim to elucidate the neural basis and underlying mechanisms driving this balance. In sum, our findings from the meta-

analytic network neuroscience approach using previously collected resting-state fMRI data were externally validated via a task-based fMRI study. Thus, our findings contribute to a concentrated cognitive and neuroscientific effort to elucidate the cognitive and neural mechanisms that realize creative thinking[2,4,41,67,83–85].

Limitations and future directions

We acknowledge several limitations in our study. First, we recognize that the "true" effect size is small, meaning that the dynamic reconfiguration between the DMN and ECN is only one factor contributing to creative ability. Based on resting-state fMRI data, our meta-analytic findings suggest that more efficient DMN-ECN interactions support higher creative ability; alternatively, as individuals develop their creative skills, that may increase the efficiency of DMN-ECN interactions, even at rest (e.g., Ref. 86). Importantly, many other factors beyond the dynamic DMN-ECN patterns observed here are certainly involved in creative ability, such as personality, attention, perception, curiosity, and creative self-beliefs. Yet, similar DMN-ECN interaction patterns have been found for other[87,88] forms of spontaneous thinking (e.g., mind wandering). Therefore, future research is needed to further explore large-scale brain networks and their dynamic interactions across different creative task contexts, to distinguish between the recruitment of ordinary and creativity-specific neural patterns.

A second limitation of our study is the application of a meta-analytic network neuroscience approach to pool raw data from previous studies. Recently, it has been proposed that extremely large sample sizes (in the tens of thousands) are needed to conduct reproducible Brain-Wide Association studies[89,90]. However, others have argued for the significance of effect size over sample size in such brain-wide association studies[91–94]. Our approach employs standard and acceptable meta-analytic methods, and the sample sizes of the included studies are adequate for network neuroscience research[95]. A recent study offers valuable insights into sample sizes and true effect sizes in brain–behavioral phenotype associations, suggesting that typical sample sizes often lead to inflated effect sizes and replication failures. However, as sample sizes increase into the thousands (e.g., $N = 2000$), replication rates improve, and effect size inflation decreases, even though the true associations may be smaller[90]. Therefore, the observed effect size in our study likely reflects the true relationship between DMN-ECN dynamic interaction and divergent thinking ability. Nonetheless, a sufficiently large sample size and a dataset with high homogeneity in data collection methods (including MRI and creativity measures) would be more beneficial for detecting this true relationship. In addition, we echo the message advanced by Grafton et al.[91] that recognizes the need for both small-sample focused studies with large-scale consortia-based studies to truly advance our understanding of how complex neural interactions realize human behavior.

Another potential limitation of the present study is that the spatial resolutions of the chosen atlases, particularly the number of regions included within the DMN and ECN, may affect the overall effect of this study. Some studies have indicated that the choice of brain atlas significantly influences the strength of the association between functional connectivity and cognitive abilities[96–98]. Therefore, future research is needed to replicate and extend our findings using additional brain atlases with different spatial resolutions.

Conclusions

The present findings reveal that highly creative people are characterized by flexible and balanced interactions between the DMN and ECN (measured at rest). Importantly, similar neural patterns were also observed during creative thinking in an external validation with task-based fMRI data. Thus, our study provides a particularly promising direction for clarifying the dynamic neural and cognitive processes that realize creative cognition. In sum, studying creative cognition from a dynamic framework holds much promise for yielding an enhanced understanding of creativity and providing new insight. Critically, our study highlights a general neural marker of creative ability, thus further demystifying the complexity of creative cognition.

Methods

Datasets

The datasets analyzed in the current study for meta-analytic were from five centers in Austria[99–102], Canada[103], China[74,104,105], Japan[28,106], and the United States[27,107], constituting multi-center datasets including structural MRI, resting-state fMRI, and creative ability data (assessed by divergent thinking tests). There were 10 samples from four centers and one open-access source, including 3533 healthy participants who completed MRI scans and 2772 participants who also completed creativity assessment. The final valid sample consisted of 2433 participants (male = 1113), with an average age of 21.12 years (SD = 4 years), after passing a set of exclusion criteria including failed fMRIprep running (due to poor spatial normalization and/or signal quality, and incomplete raw data), excessive head motion, and incomplete or invalid behavioral data. For these 10 datasets, the resting-state fMRI data (including UNCG, GBB_S1, GBB_S2, SLIM_S1, UG_S3, and TKU) have been published elsewhere[54,74,102,105–107]; the task-fMRI data collected has also been published, but the resting-fMRI data collected during the same period was not analyzed in prior work and is included in this study (e.g., QU, UG_S1, UG_S2, UG_S3). Importantly, for all the datasets assessed in the present research, no prior study has utilized the analyses conducted in the present work or examined the patterns of DMN-ECN dynamic interaction using resting-state data in relation to individual creative performance. For more details on each sample, see Supplementary Table 1. All local institutional review boards permitted the use of the de-identified and anonymized data. All study participants provided written informed consent at their local institution prior to participation. All ethical regulations relevant to human research participants were followed.

Behavioral assessment

In this study, individual creative ability was assessed by divergent thinking tasks. Despite different divergent thinking tasks employed across the ten datasets, the guidance and requirements of the task were the same, including participants being asked to come up with as many interesting, novel, and uncommon uses as possible for some objects in a limited time. Consistent with conventional scoring procedures[108], 2-4 raters scored participants' responses in each task and obtained good interrater reliability, except for the TKU sample which employed one single trained rater. For each type of divergent thinking measured in one dataset, the average of the scores for each rater was obtained, or latent variable modeling was used to model the overall creativity score. Notably, the precise scoring methods for creativity varied across different centers, encompassing measures of originality or the sum of originality and fluency (for more details on measurements and scoring, see Supplementary Dataset Information).

Image acquisition and processing

Structural and functional data were acquired locally, see more detailed scanning parameters in the Supplementary Dataset Information. All images were analyzed using fMRIPrep 1.4.1rc1 (RRID:SCR_016216)[109], which is based on Nipype 1.2.0 (RRID:SCR_002502)[110]. A full description of the preprocessing pipeline can be found in Supplementary Information —"Imaging preprocessing." After preprocessing, participants with substantial head motion (>3 mm in translation or >3° in rotation) were excluded from further analysis. Head motion was calculated after ICA-AROMA in which noise components (including motion) were identified and excluded during fmriprep processes, thus the value here is not the raw head motion. Temporal artifacts were computed in each dataset by calculating framewise displacement (FD) and FD > 0.5 mm or DVARS > 5% were identified[111]. Participants with greater than mean FD 10% of the resting time points exceeding these values were excluded from further analysis.

Dynamic functional analysis

Following preprocessing, we first computed the ROI-wise SNR via the mean magnetic resonance signal over time divided by the standard deviation of the time series, and those ROIs with inadequate signal (i.e., SNR > 2 SD above below the group mean)[112] were excluded. Subsequently, the mean time series

was extracted from regions of interest (ROIs) affiliated with two networks of interest, default mode and executive control networks, which were obtained in MNI standard space from 300 cortical parcels with 17 networks[113]. For each dataset, the temporal resolution of time-resolved connectivity in BOLD time series data of ROIs was estimated by the Multiplication of Temporal Derivatives (MTD) approach within a sliding temporal window of 14 time points, resulting in a matrix with *times* * *ROIs* * *ROIs* (Supplementary Fig. 3A). The MTD is computed by calculating the point-wise product of the temporal derivative of pairwise time series, which were reliable and replicable across multiple datasets compared to the conventional sliding-window Pearson's correlation[45]. Next, time-resolved community structure was estimated for each temporal window using the Louvain modularity algorithm from the *Brain Connectivity Toolbox*[114], in which the community assignment for each region was assessed 500 times, and a consensus partition was identified using a fine-tuning algorithm. Based on time-resolved community assignments, we estimated the within-module strength z-score (W_T) for each region and estimated the diversity of between-module strength (i.e., participation coefficient; B_T) for each region within each temporal window[115].

Using these two connectivity measures (i.e., W_T and B_T), we computed a joint histogram of within-module strength and between-module strength measures (by summing the instances of each value of the two measures within 100 equally defined bins along each axis) within each temporal window as a cartographic profile. To test whether the cartographic profile fluctuated over time, we subsequently performed k-means clustering analysis to classify each cartographic profile over time into one of two states (clusters) using a priori parameter (with k = 2 partition and 100 random restarts), which has been shown to be optimal in previous studies[45]. To determine the specific state for one cartographic profile, we compared the mean scores of regional W_T and B_T scores across the two states using an independent samples *t*-test. Here, one state with a higher B_T score and lower W_T score suggests cooperation and integration between DMN and ECN (i.e., integrated state, T_{In}), and the other state means segregation between the two networks (i.e., segregated state, T_{se}; Supplementary Fig. 3B). Referring to previous studies[45,53], we can determine which state (i.e., segregated or integrated state) dominates the functional connections between the two networks in each time window. Across the time windows, the switching frequency that measures the transition number segregated and integrated states was calculated, which was defined as: Switching frequency = $N(T_{In} \rightarrow T_{se}) + N(T_{Se} \rightarrow T_{In})$; and the balance of the two states was computed by the difference of dwell time in the two states divided by the whole time, which was defined as: Balance = $(T_{In} - T_{se})/T_{all}$, where T_{all} is the total dwell time of T_{In} and T_{se}. Previous studies have shown that frequent switching between dynamically segregated and integrated states over long time periods leads to balance. Consequently, switching frequency and balance exhibit an inverted U-shaped relationship (Supplementary Fig. 3C)[45].

Within-dataset analyses

For each dataset, we first computed the correlation between creative performance (i.e., divergent thinking ability) and switching frequency (between segregation and integration of DMN-ECN) controlling for gender, age, mean FD, and global mean signal. In addition, we calculated this relationship in each dataset using a robust correlation approach[116]. Our second question concerned whether the relationship between creative performance and balance of the two dynamic brain states is linear or quadratic. For linear and quadratic regression models, the polynomial regression model contained the balance index (i.e., linear) or the balance index and its squared term (i.e., quadratic) as independent variables, and individual creativity score as the dependent variable, as follows:

$$\text{Linear}: \ PE_{ij} = \text{Intercept} + \beta_1(\text{balance}) + u_1 + e_{ij}$$

$$\text{Quadratic}: \ PE_{ij} = \text{Intercept} + \beta_1(\text{balance}) + \beta_2(\text{balance}^2) + u_1 + e_{ij}$$

Next, ANOVA and the Akaike information criterion (AIC) were used to determine which model better fit the data if both the linear and quadratic models were significant based on the F statistic, where a smaller AIC indicated a better fit. A difference of >2 in the AIC was deemed to reflect a significant difference between models.

Meta-analytic approach

We performed meta-analytic analyses using functions available in the metafor package (version 1.9-6)[117] within the R open-source software environment (version 3.2.0)[118]. In within-dataset analyses, the *r*-value between the state-switching frequency and creative performance was calculated for each dataset. Next, the r values among the ten datasets were transformed into effect sizes using "escalc" function with the measure option set to "ZCOR". Considering that several centers included in this analysis provided multiple datasets as different projects and creativity measurements, we first used random-effects models at the location level with the "rma.mv" function with the method option set to "REML" implemented in metafor R package. Meanwhile, the random-effects model also fit using "rma" function, which also provided conventional output on the Q test for heterogeneity. Further moderator and subgroup analyses were performed to explore the potential impact of moderators, including locations, MRI protocol, scoring methods, sample size, age, and gender proportion.

Reproducibility

Beyond the meta-analytic network neuroscience approach, we also performed mega-analyses by pooling all available individual-level data from all datasets[119]. This approach allowed us to exclude non-relevant covariates, such as demographic and location factors, and to focus on specific effects. To enable this analysis, all creative performance, switching frequency and functional balance between segregated and integrated DMN-ECN states were converted to Z-scores within each dataset. Initially, in the mega-analysis, we examined the relationship between creative performance and the frequency of switching between segregation and integration within the DMN-ECN. In this analysis, the switching frequency across datasets was treated as a random factor, while the frequency of switching and non-relevant covariates were considered as fixed factors. Subsequently, we explored two models: the linear and nonlinear relationships between creative performance and the balance of segregation and integration in the DMN-ECN. We then conducted an ANOVA test and compared the AIC values between the linear and quadratic models within the mega-analysis. All mega-analyses were conducted using a linear mixed-effects model (LME), employing the 'nlme' package in R (version 4.2.3). In addition, we reanalyzed the data using the Shen brain atlas[66], which consists of 54 regions of interest within DMN and ECN to confirm our results.

External validation

The use of resting-state fMRI approaches for deriving cognitive explanations has been criticized by some[120]. In this context, event-related fMRI approaches offer, to some extent, a more direct test of creative mental operations[121]. Therefore, we conducted an external validation analysis to assess whether the dynamic DMN-ECN switching pattern could be validated in the context of a divergent thinking task involving creative cognitive processes in contrast to a control task requiring common cognitions (e.g., memory retrieval). Thirty-one participants completed both the Alternate Uses Task (AUT) and the Object Characteristics Task (OCT) during fMRI scanning. In the statistical analysis, we initially employed a paired-sample *t*-test to examine whether there was a significant difference in switching frequency between AUT and OCT tasks. Subsequently, a mixed linear or nonlinear model was used to investigate the relationship between creative performance (i.e., AUT performance) and switching frequency and balance of the two dynamic brain states.

Statistics and reproducibility

The sample size details for each dataset, as well as the statistical analyses for both behavioral data and fMRI data, are provided in the respective sections of the "Results" and "Methods."

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The processed data generated in this study are available on the Open Science Framework at https://osf.io/83arw/, https://doi.org/10.17605/OSF.IO/83ARW. Some of the raw data used in the present study has been publicly available and analyzed in previous studies, and a part of the non-public data is available from the corresponding author upon reasonable request.

Code availability

The code generated for this study has been shared on the Open Science Framework at https://osf.io/83arw/, https://doi.org/10.17605/OSF.IO/83ARW.

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Author contributions

Q.C., Y.N.K., and R.E.B. conceptualized the study; H.T., A.F., M.B., R.K., J.Q., and R.E.B. contributed the data; Q.C., Y.N.K., Z.C., K.Z., and R.E.B. performed the analyses; Q.C., Y.N.K., and R.E.B. wrote the original draft; Z.C., M.B., D.C.Z., K.Z., J.L.C., J.Q., and R.E.B. review and editing. All co-authors provided feedback and approved the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Yoed N. Kenett or Jiang Qiu.

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